SP-RAN: Self-Paced Residual Aggregated Network for Solar Panel Mapping in Weakly Labeled Aerial Images

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***Abstract*— With the rapid development of the solar distribution, solar panel mapping is becoming increasingly valuable to decision-makers. Weakly supervised methods have been developed to reduce the cost in training sample collection, and the most successful ones follow the alternative training scheme, which first generates coarse object localizations as pseudo labels (PLs) and then utilizes these PLs to train an end-to-end network for object extraction. As remote sensing images are typically characterized by multiple occurrences of objects and complicated backgrounds, the alternative training scheme suffers from low mapping accuracy and deficient boundary maintenance due to the varying quality of PLs. In this article, we focus on addressing these problems by adaptively adjusting the contributions of quality-varying PLs and propose a novel self-paced residual aggregated network (SP-RAN) for solar panel mapping. Specifically, with the initial PLs generated by gradient-weighted class activation mapping, a residual aggregated network is designed for target mapping with special consideration for the capability in producing complete and well-shaped mapping results. Considering the inconsistent quality of PLs, an effective confidence-aware (CA) loss is developed to emphasize the contribution of high-quality PLs and alleviate the negative impacts brought by the bad-quality ones in the training phase. Moreover, to concentrate on boundary maintenance, a novel self-paced label correction (SP-LC) strategy is proposed to selectively update PLs by considering their reliability. Extensive experimental comparisons with state-of-the-art methods and ablation study on two aerial datasets and a remote sensing dataset demonstrate the superiority of the proposed method.**

***Index Terms*— Object mapping, remote sensing, residual aggregated network (RAN), self-paced learning, weakly supervised learning (WSL).**

I. Introduction

As the most popular and dominant type of solar power technology, solar photovoltaic (PV) sees rapid growth globally in the past decades [1]. Despite the environmental and economic benefits, the significant increase in solar PV deployment, especially small-scale solar installations, influences the electricity grid's behavior greatly due to the presence of varying loads and network conditions [2], [3]. Monitoring solar deployment, including locations, sizes, and power capacity, not only helps minimize safety risks but also aids the government and electricity companies in gaining better insights into solar energy utilization.

Automatic solar panel mapping has been investigated in recent years, mostly by applying machine learning algorithms to high-resolution remote sensing images (RSIs). Early works followed the general rules in supervised object extraction [4]–[6], which consist of two main phases: 1) data representation and feature learning and 2) classifier selection and training. Handcrafted features' descriptors, including local color statistics [5] and the Gabor energy filter [7], were utilized to produce feature representation, and then, traditional classifiers, such as support vector machine (SVM) [4] and random forests [5], were trained with the generated feature vectors to locate solar panels with sliding windows. The discriminative power of specially designed features and classifiers plays a key role in the success of these approaches.

Nowadays, convolutional neural networks (CNNs), as one of the most prevalent deep learning techniques [8], have achieved promising results in the field of solar panel mapping. Malof *et al.* [9] utilized the VGG net [10] as a detector to determine the existence of solar panels in small square patches. Yuan *et al.* [11] considered solar panel mapping as a semantic segmentation task and proposed a CNN-based method to produce pixelwise classification. Later, modified versions of classical deep network architectures [12]–[14] were proposed to generate dense mapping results. Wang *et al.* [12] introduced SegNet [15] to produce dense predictions of PV arrays' locations and sizes. Similarly, Castello *et al.* [13] applied a CNN in the shape of U-net [16] to discover the rooftop solar installations. Some researchers observed that traditional machine learning algorithms and CNN-based approaches have complementary advantages in identifying true positives and true negatives, respectively, and attempted to develop hybrid methods for accurate identification [17], [18].

Most existing CNN-based mapping methods require training samples with accurate pixelwise annotations in the training phase, which is prohibitively human labor-intensive, especially in remote sensing tasks. To reduce the labeling workload, weakly supervised learning (WSL) [19] resorts to weak annotations for supervision, such as image-level tags and scribble-based labels, which are much easier to obtain. WSL-based methods, however, are very challenging

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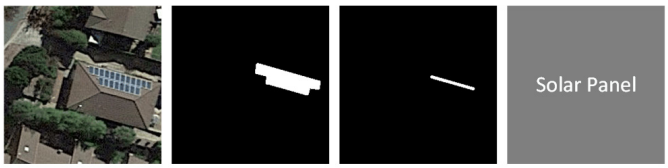


Fig. 1. Different levels of supervision. (a) Training sample. (b) Pixelwise labels. (c) Scribble-based labels. (d) Image-level labels.

as accurate location and shape cues are expected to be inferred from weak annotations, as shown in Fig. 1(c) and (d). They usually suffer from missing objects, ambiguous outlines of the detected objects, and incomplete or over prediction. To minimize the gap between weakly and fully supervision, most existing methods leveraged an alternative training scheme [20]–[24]: weakly supervised object localization methods, such as class activate mapping (CAM) [25] and gradient-weighted CAM (Grad-CAM) [26], are first utilized to discover discriminative but coarse regions of the objects; then, with generated regions employed as "fake" labels, i.e., pseudo labels (PLs), end-to-end networks are trained to discover the objects in completeness. Saliency cues [24], [27], boundary information [23], [28], and conditional random field (CRF)-based consistency [29], [30] were also exploited as additional information to help find the whole object extent. To take advantage of dynamic supervision, label updating strategy [20], [31] was also incorporated to find sharp edges for the detected objects. Another promising direction is to tackle the WSL problem from the perspective of label noise [32]. Most methods belonging to this family focus on designing noise-robust loss [33], reweighting training samples [34], and correcting bad samples [35].

For RSIs, learning from weakly annotated samples can be more difficult since RSIs are typically characterized by a sparser distribution of objects, ambiguous boundaries due to the limited spatial resolution, and complicated background. Considering these characteristics, insightful works [36]–[41] have been proposed in the literature to reduce background interference and improve accuracy. Recently, WSL-based methods with image-level annotations have been intensively investigated [42]–[45] since the image-level label is one of the most efficient types of weak labels, indicating the existence of objects belonging to a certain class in the image. Hierarchical feature representation [42], curriculum learning [45], [46], and attention mechanism [43] have been actively studied and combined with the alternative training scheme to further improve the performance. Without any supervision of actual panel outlines, Yu *et al.* [47] developed the first weakly supervised method specially designed for solar panel mapping, named "DeepSolar," in which Inception-v3 [48] and CAM are incorporated to obtain integral objects in completeness. Zhang *et al.* [31] proposed to generate PLs with Grad-CAM [26], and a novel label correction strategy was proposed to refine the mapping results iteratively.

Despite the efforts made in improving the performance, the alternative training scheme faces two problems. The first one is how to avoid the negative impacts resulted from the bad-quality PLs given by the weakly supervised object localization. Currently, most existing methods rely on CAM and Grad-CAM to produce the initial localization maps, which only correspond to the most discriminative part of the objects. Inaccuracy of such PLs misleads the model training as the network optimizes over mislabeled pixels [33], leading to the unsatisfactory performance in mapping accuracy. Particularly, the quality of PLs varies greatly as the background changes [49], [50]. Due to the inability of CAM and Grad-CAM to locate multiple instances appearing in an image, the situation would be worse, and bad-quality PLs would do great harm to the model training. The second problem is how to avoid partial prediction and produce the mapping results with precise shapes. Due to the coarse PLs employed, weakly supervised methods generally predict the most discriminative object regions and present a weak ability to maintain well-defined boundaries. Label updating has been revealed as an effective way to improve object completeness [51], [52] and boundary maintenance [24] via making the PLs approach the actual object outlines progressively. However, most existing label updating strategies update all the PLs in the training phase and omit the varying quality of the initial PLs, which may introduce new falsely labeled pixels and lead to error accumulation.

To address these problems, we propose a self-paced residual aggregated network (SP-RAN) for weakly supervised solar panel mapping by using a limited number of training data at the image level and a large number of unlabeled data. Instead of blindly trusting the initial PLs, SP-RAN focuses on emphasizing the contribution of high-quality PLs and reducing negative impacts brought by the bad-quality ones in the training phase. As illustrated in Fig. 2, the proposed method starts from self-pixel labeling to generate initial PLs with an image-level classification network. Then, treating these labels as training samples with uncertainty, a residual aggregated network (RAN) with an effective confidence-aware (CA) loss is trained with a proposed self-paced label correction (SP-LC) strategy to selectively update the PLs and boost the model performance progressively. Moreover, a large number of unlabeled data are automatically labeled by the self-pixel labeling process and added into the model training to improve the generalization ability. The well-trained RAN is then used to produce the mapping results. The main contributions of this article are summarized as follows.

1) To avoid partial prediction, an effective CA loss is proposed to train the RAN by dynamically adjusting the contribution of PLs with varying qualities. We propose to quantify the quality of PLs by employing the confidence given by the image-level classification network. Taking the confidence as the weights, the CA loss dynamically suppresses the effects of training samples with corrupted PLs and emphasizes the contributions of samples with relatively clean PLs.
2) A novel SP-LC strategy is designed to minimize the negative effect from irrelevant surroundings when approaching sharp object outlines during the iterative training. Instead of updating the entire PLs set at once, the SP-LC strategy focuses on updating PLs in a self-paced manner:

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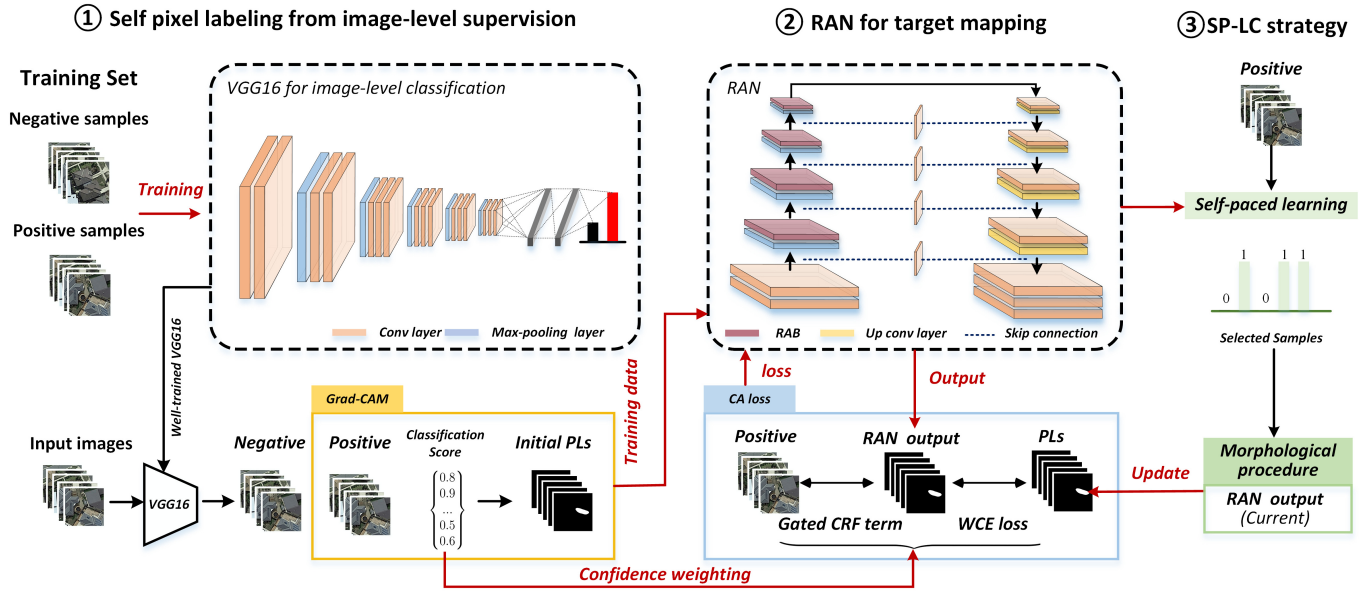


Fig. 2. Flowchart of the SP-RAN.

the least ambiguous PLs will be updated first, and the rest will be considered progressively. Selected PLs at each round will be updated by combining the current output with a morphological procedure to minimize the error accumulation in the iteration. With progressively corrected labels, the model becomes increasingly reliable and approaches actual object outlines.

3) The design of RAN considers the merits of feature pyramid network (FPN) and residual aggregated blocks (RABs). FPN is adopted as the backbone as it has a significant advantage in localizing varying-size objects accurately by fully utilizing multiscale feature representation. In the bottom-up path of the RAN, the RAB is further incorporated to preserve abundant spatial details for outline recovery.

The remainder of this article is organized as follows. In Section II, we introduce the SP-RAN. Experimental results are reported in Section III. Finally, the conclusion is given in Section IV.

II. Methodology

In this section, the proposed SP-RAN will be presented in detail. As shown in Fig. 2, the SP-RAN is composed of three steps: self-pixel labeling from image-level supervision, RAN for target mapping, and SP-LC strategy. The purpose of the first step is to generate initial PLs from image-level annotations with the aid of a well-trained classification network and Grad-CAM. Then, the second step aims to produce initial mapping results with the proposed RAN trained by the CA loss function. In the last step, the SP-LC strategy is proposed to selectively update PLs in a self-paced manner. Unlabeled data are also labeled and added into model training.

A. Self-Pixel Labeling From Image-Level Supervision

To obtain pixelwise labels from image-level annotations, we utilize an image-level classification network and

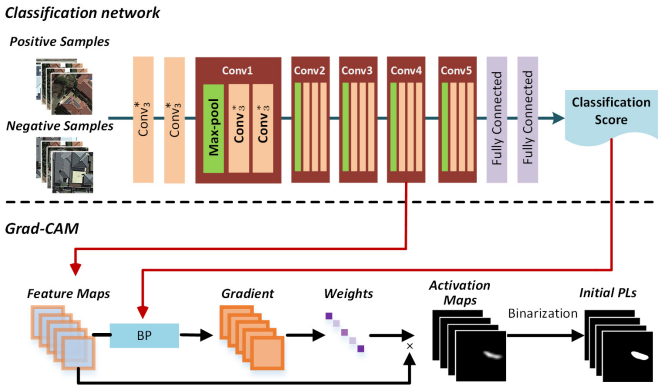


Fig. 3. Flowchart of self-pixelwise label generation from image-level supervision.

Grad-CAM to provide initial object localization. We first adopt the VGG16 net as the backbone to train a classification network, and the Grad-CAM is then leveraged to provide class activation maps, which are subsequently transformed into binary masks via Otsu thresholding. Fig. 3 shows the flowchart of label generation from image-level supervision.

In our work, the provided imagewise label indicates whether or not there are targets in the image (positive or negative samples). Then, the classification network is trained to solve a binary classifications problem. VGG16 net [10] was proposed by Simonyan *et al.*, which contains 13 convolutional layers, five max-pooling layers, three fully connected (FC) layers, and, finally, a soft-max layer. All the convolutional layers have a small 3×3 receptive field, 1-pixel stride, and 1-pixel padding, followed by a rectified linear unit (Relu) to increase nonlinearity. Each max-pooling layer with stride 2 is carried out over a 2×2 window, separating the convolutional layers into five blocks. For the classification network, we alter the FC layers in the VGG16 net to 256 and two channels before the soft-max activation to make it suitable for the binary

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classification problem. The 2-D output of the classification network is the probabilities of the input sample belonging to positive or negative cases, respectively. The cross-entropy loss is utilized to train the classification network. Here, we use $\mathbf{F}_{i,q}^{k}$ to denote the kth channel in the feature maps produced by the ith convolutional layer of the qth block in the classification network.

Then, Grad-CAM takes the feature maps produced by the third convolutional layer in the fourth block in the classification network as the input to generate class activation maps. Denote the rectified linear unit as Relu. The gradient of the classification score $f(\mathbf{I};\Theta)_c$ for class c to $\mathbf{F}_{i,q}^{k}$ is first calculated, and the global average pooling, i.e., GAP, is subsequently used to generate the weights, as given in (1). After each channel's feature map is weighted by its weight $\widehat{w}_{i,q,c}^{k}$, the sum leads to the final class activation map $\mathbf{M}_{i,q,c}$ [as shown in (2)]

$$\widehat{w}_{i,q,c}^{k} = \mathrm{GAP}\left[\frac{\partial f(\mathbf{I};\Theta)_c}{\partial \mathbf{F}_{i,q}^{k}}\right] \tag{1}$$

$$\mathbf{M}_{i,q,c} = \mathrm{Relu}\left\{\sum_{k=1} \widehat{w}_{i,q,c}^{k} \cdot \mathbf{F}_{i,q}^{k}\right\} \tag{2}$$

where $\mathbf{I} \in \mathbf{R}^{H\times W\times C}$ is the input image with the size of $H \times W \times C$, and Θ represents the parameter set of the classification network.

After obtaining the class activation map for the foreground class, i.e., solar panels, the Otsu segmentation method is used to produce binary masks that are subsequently taken as initial PLs in the target mapping network.

B. Residual Aggregated Network for Target Mapping

When alleviating the reliance on imagewise annotations, weakly supervised object segmentation has three major challenges: 1) missing objects due to large appearance variations; 2) ambiguous outlines of the detected objects; and 3) incomplete prediction or over prediction. The design of RAN is to meet these challenges. To address the first challenge, we adopt FPN as the backbone of the target mapping network, as FPN has a significant advantage in localizing varying-size objects accurately by fully utilizing multiscale feature representation [53]. However, supervised by coarse and inaccurate PLs, FPN exhibits a deficient ability to avoid partial prediction and preserve object outlines. To tackle the second challenge, RABs are incorporated with FPN to preserve abundant spatial details for precise outline recovery. Finally, to extract the object as a whole, we propose a new CA loss, which improves the object completeness via suppressing the negative influence brought by the training samples with low-quality PLs. The architectures of the proposed RAN and RAB are presented in Fig. 4.

1) Residual Aggregated Block: If a deep CNN is built by stacking consecutive convolutional layers or multiple residual blocks (RBs), features produced by the former layer are passed to the deeper layers sequentially, which means that local features are generated by preceding layers or blocks can hardly propagate to the end of the network. Hence, local information that needs to be preserved makes very limited contributions to the training process of the entire network and the final output, which inevitably leads to performance degradation. This is particularly the case for tasks requiring well-preserved spatial details and object outlines.

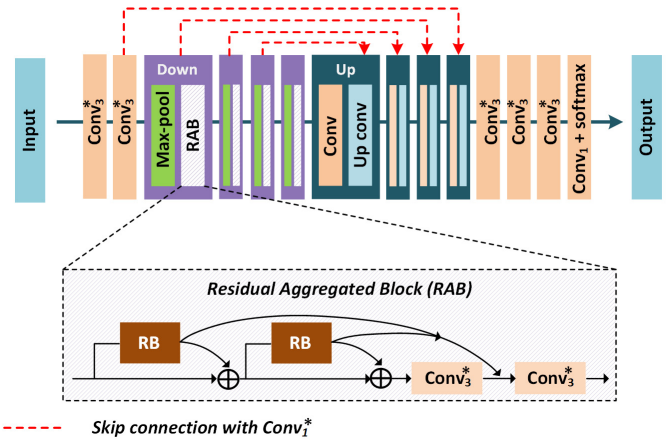


Fig. 4. Flowchart of the proposed RAB and RAN for target mapping.

The RAB is motivated by the residual feature aggregation framework [54], which is originally proposed for image super-resolution. Similar to the residual feature aggregation framework, RAB makes the most of the local residual features by propagating low-level residuals of all the proceeding RBs to the end of the block. As shown in Fig. 4, each RAB contains two stacked RBs and two convolutional layers with 3×3 kernel and 1-pixel stride. The residual features of the two RBs are concatenated directly with the output of the second RB after the first convolutional layer and then sent to the last convolutional layer, after which the channel of the feature maps doubles in number.

Denote the input and output of the jth RB in RAB as $\mathbf{R}_j^{\mathrm{in}}$, and $\mathbf{R}_j^{\mathrm{out}}$, where $j = 1, 2$. The output of the first RB is sent to the second RB as the input. The output of the RAB can be calculated as follows:

$$\mathbf{Q}_j = \mathrm{Conv}_3\left(\mathrm{Conv}_3^*\left(\mathbf{R}_j^{\mathrm{in}}\right)\right), \quad j = 1, 2 \tag{3}$$

$$\mathbf{R}_j^{\mathrm{out}} = \mathbf{Q}_j + \mathbf{R}_j^{\mathrm{in}} \tag{4}$$

$$\mathbf{B}^{\mathrm{out}} = \mathrm{Conv}_3^*\left(\mathrm{Concate}\left[\mathbf{Q}_1, \mathbf{Q}_2, \mathrm{Conv}_3^*\left(\mathbf{R}_2^{\mathrm{out}}\right)\right]\right) \tag{5}$$

where Conv_3^* and Conv_3 represent a single 3×3 convolutional layer with and without Relu, respectively. $\mathbf{Q}_j$ is the residual feature generate by the jth RB in RAB. Concate$[\cdot]$ denotes the concatenate operation.

2) Residual Aggregated Network: Following the FPN architecture, RAN is constructed in an encoder and decoder manner, with a shallow feature extraction module, a bottom-up path, and a top-down path with skip connections linking the two paths. In the bottom-up path, four RABs are stacked, with a max-pooling operation applied before each block. In the top-down path, coarse but semantically strong feature maps are first upsampled and then merged with high-resolution but low-level feature maps generated by the corresponding blocks in the bottom-up path via skip connections. The network architecture of RAN is detailed as follows.

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Given the input image $\mathbf{I}$, two separate 3×3 convolutional layers with each followed by the Relu are first utilized to produce the initial features $\mathbf{F}_0$, which is then sent to the bottom-up path

$$\mathbf{F}_0 = \mathrm{Conv}_3^*\big(\mathrm{Conv}_3^*(\mathbf{I})\big). \tag{6}$$

The bottom-up path is composed of four repeated applications of 2×2 max-pooling operation for downsampling and RAB blocks. At each RAB block, we double the number of feature channels. The output of $(p-1)$th RAB is first downsampled by a 2×2 max-pooling operation and then fed to the pth RAB as the input. The output of the pth RAB can be written as follows:

$$\mathbf{B}_p^{\mathrm{out}} = \mathcal{R}_p\left[\downarrow \mathbf{B}_{p-1}^{\mathrm{out}}\right] \tag{7}$$

where $\downarrow$ denotes the 2×2 max-pooling operation. $\mathcal{R}_p[\cdot]$ denotes the feedforward propagation of the pth RAB block.

In the top-down path, every step contains a 3×3 convolutional layer with 1-pixel stride, followed by the Relu, and a 2×2 up convolutional layer (up conv layer) with a 2-pixel stride that halves the number of feature channels. To enhance high-level features with high-resolution features, skip connections are utilized to make elementwise addition of corresponding feature maps from the bottom-up and top-down paths. Before the addition, feature maps from the bottom-up path undergo a 1×1 convolutional layer to reduce channel dimensions. Three extra successive 3×3 convolutional layers with 1-pixel stride are appended to reduce the aliasing effect of upsampling operation and pixelwise addition. Finally, a 1×1 convolutional layer with soft-max activation is added to produce the final results.

3) Confidence-Aware Loss: Although Grad-CAM provides reasonable object location cues, it has a limited ability to capture multioccurrence of objects and usually presents incomplete and over prediction. Training target mapping network with quality-varying PLs may lead to severe confusion and produce partial prediction only covering the most discriminative regions. Thus, how to measure the quality of PLs and emphasize the contributions of high-quality ones is crucial. Instead of only selecting high-quality samples for training and omitting bad-quality ones, we propose a CA loss and adjust the contribution based on the quality scores produced by the image-level classification network. Specifically, the CA loss is composed of three parts: weighted cross-entropy, the gated CRF regularizer, and confidence weighting.

a) Weighted cross-entropy: Generally, solar panels belong to the minor class compared to other major urban classes in the background of aerial images. To mitigate the influence of the imbalance between the foreground and background, we use weighted cross-entropy L_{wce} instead of cross-entropy as the first term in the CA loss, as shown in the following equations:

$$L_{\mathrm{wce}} = -\sum(\alpha \cdot \mathbf{G} \cdot \log(\mathbf{f}_0) + (1-\alpha) \cdot (1-\mathbf{G}) \cdot \log(\mathbf{f}_1)) \tag{8}$$

$$\alpha = \frac{N - N_{\mathrm{fore}}}{N} \tag{9}$$

where N_{fore} is the number of pixels belonging to the foreground in the ground truth $\mathbf{G}$. N is the total number of pixels in the images. $\mathbf{f}_0$ and $\mathbf{f}_1$ are the first and second channels in the final output, respectively.

b) Gated CRF regularizer: To make the mapping network concentrate on the object boundaries with background noise, the gated CRF loss [30] is introduced as the regularizer term. Inspired by the classical CRF model, the gated CRF loss calculates the pairwise energy term $\psi_{x,y}$ in a $r \times r$ local neighborhood window Ω_r, which measures the featurewise similarity with pairwise potential $k(\cdot,\cdot)$

$$\psi_{x,y}(\mathbf{f}) = \mu\big(\mathbf{f}_x, \mathbf{f}_y\big) k\big(\mathbf{V}_x, \mathbf{V}_y\big) \tag{10}$$

$$\mu\big(\mathbf{f}_x, \mathbf{f}_y\big) = \begin{cases} 1, & \text{if } \mathbf{f}_x = \mathbf{f}_y \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

where μ is the class compatibility matrix. $\mathbf{f}$ is the output of the mapping network. $\mathbf{f}_x$ and $\mathbf{f}_y$ are the outputs at positions x and y, respectively. $\mathbf{V}_x$ and $\mathbf{V}_y$ are the feature vectors for pixels at positions x and y, respectively.

In this article, we adopt Pott's class compatibility model and the contrast-sensitive two-kernel potential, which combines color feature and spatial feature into a 5-D feature vector for color-proximity similarity. The two-kernel potential utilized in this work is shown as follows:

$$k\big(\mathbf{V}_x, \mathbf{V}_y\big) = w^{(1)} \exp\left(-\frac{|\mathbf{P}_x - \mathbf{P}_y|^2}{2\theta_1^2} - \frac{|\mathbf{S}_x - \mathbf{S}_y|^2}{2\theta_2^2}\right) + w^{(2)} \exp\left(-\frac{|\mathbf{P}_x - \mathbf{P}_y|^2}{2\theta_3^2}\right) \tag{12}$$

where $\mathbf{P}_x$ and $\mathbf{P}_y$ denote the spatial positions of locations x and y, respectively. $\mathbf{S}_x$ and $\mathbf{S}_y$ denote the spatial feature vectors in locations x and y, respectively. $w^{(1)}$ and $w^{(2)}$ are the weights for controlling the influence of the two kernels. θ_1, θ_2, and θ_3 control the degrees of nearness and similarity.

After measuring the total energy over Ω_r, L_{crf} can be obtained

$$L_{\mathrm{crf}} = \sum_x \sum_{y \in \Omega_r(x) \backslash x} \psi_{x,y}(\mathbf{f}). \tag{13}$$

c) Confidence weighting: To adjust the contribution of PLs with varying qualities, we employ the classification score given by the classification network as the confidence. Training samples with high confidence are supposed to contain clear and easy-to-learn objects, while those with low confidence are more like to contain ambiguous and hard-to-discriminate targets. After weighting the WCE loss and the gated CRF regularizer term with the classification score, the CA loss can be computed as follows:

$$L = \sum_m f(\mathbf{I}; \Theta)_{c=1}^{(m)} \cdot \left(L_{\mathrm{wce}}^{(m)} + \lambda L_{\mathrm{crf}}^{(m)}\right) \tag{14}$$

where $L_{\mathrm{wce}}^{(m)}$ and $L_{\mathrm{crf}}^{(m)}$ are the WCE loss and the gated CRF regularizer for the mth training sample. $f(\mathbf{I}; \Theta)_{c=1}^{(m)}$ is the classification score of the foreground class, i.e., $c = 1$ for the mth training sample. λ balances the WCE loss and the gated CRF regularizer term.

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C. Self-Paced Label Correction

Compared with training the mapping network with fixed PLs, supervision with progressively corrected PLs has been revealed more beneficial to model training. Although some criteria based on CRF, region growing, and so on have been incorporated, label updating operation, however, may cause error accumulation as it may introduce false-labeled pixels. Hence, how to avoid over correction or under correction has been a problem worthy of attention. As solar panels usually have similar colors with rooftops and exhibit ambiguous edges, resorting to CRF, saliency cues or boundary information may not be helpful in this case.

Motivated by the observations, we propose an effective SP-LC strategy, which first utilizes self-paced learning to select training samples, whose PLs are close enough to the output of the currently trained model. These PLs are considered high-quality and reliable ones. Then, the PLs of the selected training samples are corrected by processing the network output with a morphological procedure. With a gradually improved threshold for self-paced learning and model capability, a growing number of training samples and corresponding PLs will be selected and corrected, which will help develop a reliable model with enough robustness against noisy PLs. Despite sharing a similar idea of label update with Ren's work [51] and Lin's work [52], the proposed SP-LC module has two distinct features: instead of discovering missing proposals at the instance level, SP-LC operates at an image level and is designed to alleviate the potential errors, i.e., newly introduced false "labels" resulted from the label correction operation via considering the degree of confidence and the level of quality of the samples; furthermore, the SP-LC module does not require any changes to the network architecture, so it can be easily plugged into any backbone networks during training.

Self-paced learning [55], [56] is a recently proposed learning paradigm aiming to start the learning process with easy samples and then gradually add hard samples into training. Sharing a similar concept to curriculum learning [57], self-paced learning adjusts the learning pace according to the model ability by adding a new regularization term to the learning objective, shown as follows:

$$\min_{\Phi,\mathbf{v}\in[0,1]^n} \mathbb{E}(\Phi,\mathbf{v};\delta)=\sum_{m=1}^{n} v_m L\left(\mathbf{G}^{(m)}, g\left(\mathbf{I}^{(m)};\Phi\right)\right)+f(\mathbf{v},\delta) \tag{15}$$

where $L(\cdot,\cdot)$ is the loss function for the mapping network. $g(\mathbf{I}^{(m)};\Phi)$ denotes the output of the mapping network with parameter set Φ for the mth training sample. $\mathbf{G}^{(m)}$ is the label for the mth training sample. $\mathbf{v}=[v_1, v_2, \ldots, v_n]$ is the weights indicating the training samples' learning difficulties. n is the total number of training samples.

$f(\mathbf{v},\delta)$ is the self-paced regularizer controlling the selection of the training samples according to the threshold δ. In this article, we employ a simple but effective l_1-norm regularizer, which indicates that samples with smaller loss are viewed as "easy" samples and, thus, should be processed first

$$f(\mathbf{v},\delta)=-\delta\sum_{m=1}^{n} v_m. \tag{16}$$

The optimization of (15) and (16) can be formulated as an alternative convex search (ACS) problem [58], in which variables are first classified into two blocks. Then, with one block fixed, the other block of variables can be optimized. This operation is conducted in an iterative way. With the network parameter set Φ fixed, the optimal value for $\mathbf{v}=[v_1, v_2, \ldots, v_n]$ can be obtained by

$$v_m=\begin{cases}1, & L\left(\mathbf{G}^{(m)}, g\left(\mathbf{I}^{(m)};\Phi\right)\right)<\delta\\ 0, & \text{otherwise.}\end{cases} \tag{17}$$

In each epoch, δ helps determine which training samples are allowed to participate in the label correction procedure. Denote the epoch as t, the threshold for t epoch as $\delta^{(t)}$, and the set of selected training samples as $\mathbf{S}^{(t)}$. In the set, $\mathbf{S}^{(t)}$, $\mathbf{G}^{(t,m)}$, and $\mathbf{f}^{(t,m)}$ represent corresponding PLs and the output of the mapping network of the mth training samples, respectively. For the selected training samples in the $t+1$ epoch, the output of the current mapping network is first binarized by the Otsu thresholding and then undergoes an opening operation with 9×9 structuring element **SE**, which can help reduce the false positives in the training

$$\text{Mask}^{(t,m)}=\text{otsu}\left(\mathbf{f}^{(t,m)}\right) \tag{18}$$

$$\mathbf{G}^{(t+1,m)}=\text{Mask}^{(t,m)}\circ\mathbf{SE}=\left(\text{Mask}^{(t,m)}\oplus\mathbf{SE}\right)\ominus\mathbf{SE} \tag{19}$$

where $\circ$, $\oplus$, and $\ominus$ denote the opening, dilation, and erosion operation, respectively.

As the training phase goes, we relax the threshold; thus, a growing number of training samples with higher learning difficulties will be gradually appended into the label correction process

$$\delta^{(t)}=\delta^{(0)}\cdot\exp\left(\frac{t}{\sigma}\right) \tag{20}$$

where $\delta^{(0)}$ is the predetermined threshold and σ controls the pace of the growing difficulty.

The proposed SP-LC strategy is summarized in Algorithm 1.

III. Experiments

A. Datasets

To reveal the validity of the proposed method, we evaluate the performance of SP-RAN on two aerial datasets, i.e., GMS-ACT and GMS-BRIS, both of which were obtained with Google Static Map API. The GMS-ACT dataset was collected in Canberra, ACT, Australia, with the spatial resolution varying from 0.15 to 0.3 m. Regarding the training samples with image-level annotations, the training set of GMS-ACT contains 437 positive samples and 414 negative samples, while the testing set consists of 98 positive samples and 92 negative samples. The GMS-ACT dataset is also equipped with 3405 unlabeled samples. Images in the GMS-BRIS dataset were captured in Brisbane, QLD, Australia. In the GSM-BRIS, 457 positive samples and 676 negative samples are used for the training; 99 positive samples and 117 negative ones are employed for testing. The unlabeled data are composed of 5588 samples. All the images employed have the size of 256×256 pixels. Detailed information for the two datasets is summarized in Table I.

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TABLE I
DETAILS OF THE GSM-ACT AND GMS-BRIS DATASETS USED FOR EXPERIMENTS

Data set	Location	Resolution	Band	Size	Training set		Testing set		Unlabeled data
					Positive	Negative	Positive	Negative	
GSM-ACT	Canberra	0.15-0.3 m	RGB	256×256	437	414	98	92	3405
GSM-BRIS	Brisbane	0.3 m	RGB	256×256	457	676	99	117	5588

Algorithm 1: SP-LC Strategy

Data: Input dataset $\mathscr{D}$;
Initial pseudo labels by Grad-CAM $\mathbf{G}^{(0)}$;
Initial threshold $\delta^{(0)}$;
Pre-trained mapping network with parameter set $\Phi^{(0)}$
Result: Network parameter set Φ.

```
Initialize v as v^(0) ;
t ← 0 ;
while not converged do
    Calculate δ^(t+1) by Eq. (20);
    v^(t+1) ← argmin_v E(Φ^(t), v; δ^(t+1));
    if v_m^(t+1) then
        Update G^(t+1,m) by Eqs. (18) and (19);
    end
    Φ^(t+1) ← argmin_Φ E(Φ, v^(t+1); δ^(t+1));
    t ← t + 1;
end
Return Network parameter set Φ;
```

B. Implementation Details

The image-level classification network and the target mapping network are trained separately in the training phase. The classification network based on VGG16 is utilized by Grad-CAM to produce initial PLs. The well-trained VGG16 is also used to produce classification scores for the CA loss, which is subsequently employed to train the target mapping network. In the testing phase, only the target mapping network is required.

For the classification network, we use the training set with image-level annotations, which reflects the existence of desired objects in an image. VGG16 is adopted as the backbone. As it is difficult to train VGG16 from scratch, we fine-tuned the pretrained VGG16, with all the convolutional layers' weights initialized with the pretrained model over the ImageNet datasets [59]. The weights of three FC layers are randomly initialized with a truncated normal distribution, where mean and standard deviation are set to 0 and 0.01, respectively. For both datasets, we use cross-entropy to train the classification network with the RMSProp optimizer. The batch size is 16, and the learning rate is 10^{-4}. The training phase ends after 10 epochs.

The training of the target mapping network is composed of two stages: initial training with the training set and incremental training with SP-LC and unlabeled data. Both training procedures use the Adam optimizer with a batch size of 8. In the initial training, the initial learning rate is 10^{-3} and decreased by a factor of 0.8 after every 30 epochs for the GMS-BRIS dataset and 0.9 after every 20 epochs for the GMS-ACT dataset. This stage terminates after 200 epochs. In the incremental training, more positive samples are identified by the classification network from the unlabeled data, and then, Grad-CAM is utilized to produce PLs for the RAN. This stage takes 100 epochs with a learning rate of 10^{-5}.

For the gated CRF regularizer term of the CA loss, we set the window size $r = 11$, $w^{(1)} = 0.9$, and $w^{(2)} = 0.1$ in the two-kernel potential $k(\cdot, \cdot)$ for both datasets. On the GMS-ACT dataset, the weight λ for balancing the WCE loss and the gated CRF regularizer term is 0.3. θ_1, θ_2, and θ_3 are set as 6, 16, and 6. On the GMS-BRIS dataset, we set $\lambda = 0.1$, $\theta_1 = 6$, $\theta_2 = 30$, and $\theta_3 = 6$, respectively.

In the SP-LC, threshold $\delta^{(0)}$ is selected according to the loss in the initial training stage. For the GSM-BRIS dataset, $\delta^{(0)} = 0.02$ and $\sigma = 200$; for the GSM-ACT dataset, $\delta^{(0)} = 0.04$ and $\sigma = 200$.

The proposed method was developed with a publicly available Tensorflow framework and executed on NVIDIA Tesla V100 GPU with 32-GB memory. The code is publicly available at https://github.com/zhangjue1993/Weakly-Supervised-Solar-Panel-Mapping.

C. Comparison With the State of the Art

We compared the proposed SP-RAN with HWSL [42], WS-SOD [28], PSL [45], Grad-CAM [26], DeepSolar [47], and PS-CNNLC [31], which are state-of-the-art weakly supervised object segmentation methods proposed recently.

HWSL integrates class-specific multiscale salient features generated by a well-trained classification network and obtains the segmentation results after a subtraction operation. WS-SOD uses an auxiliary edge detection branch and a gated structure-aware loss as constraints for the object structure and boundary. Inspired by curriculum learning, PSL first constructs a shallow feedback saliency analysis network for semantic segmentation and then adopts a new denoising loss function to mitigate the effect brought by missing judgment in initial PLs. DeepSolar extends class activation mapping with greedy layerwise training, aiming to find the precise outlines of solar panels. PS-CNNLC is a simple application of the alternative training scheme in weakly supervised solar panel mapping, in which Grad-CAM is first utilized to generate PLs and empirical constraints are employed to correct the PLs progressively. Although both PS-CNNLC and proposed SP-RAN adopt the alternative training scheme, SP-RAN is superior in the following aspects: first, SP-RAN resorts to

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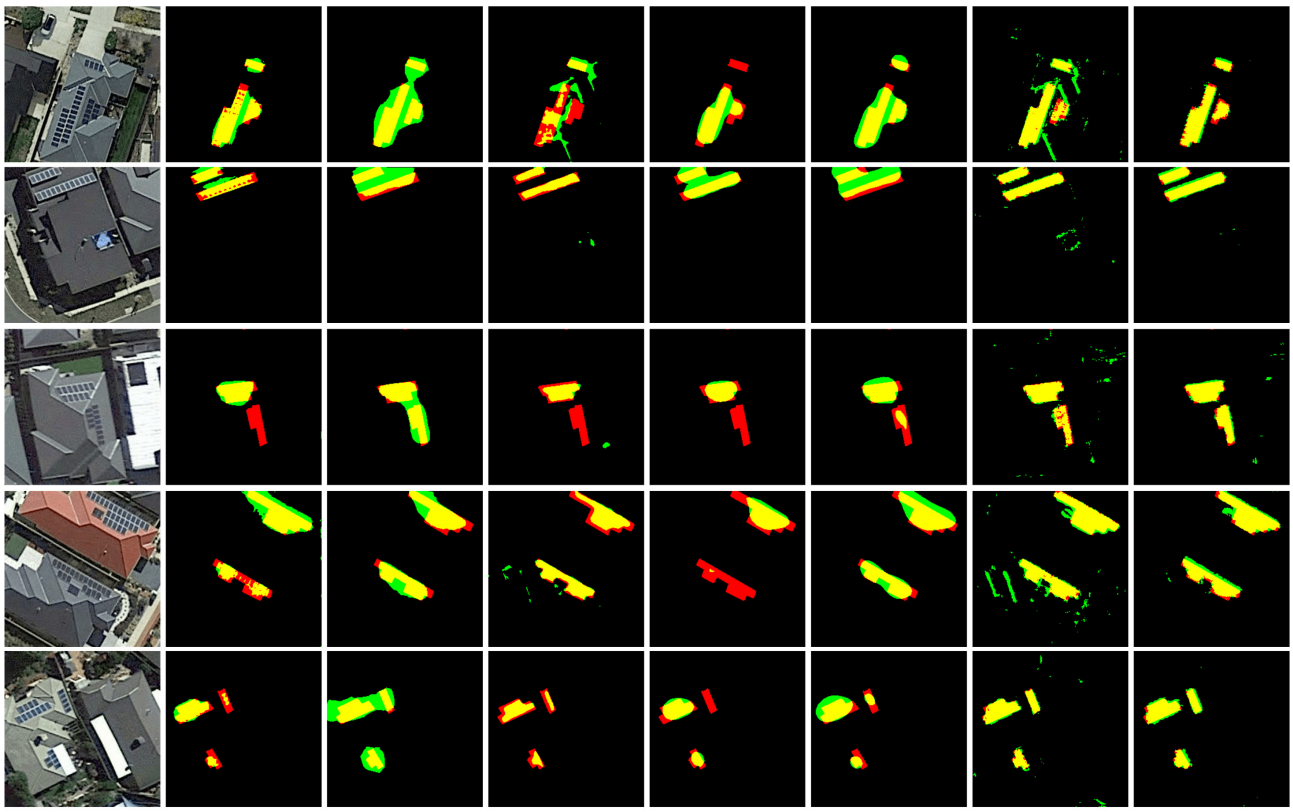


Fig. 5. Comparisons of mapping results on the GMS-ACT dataset, marked with true positives (yellow), true negatives (black), false positives (green), and false negatives (red). (a) Original images. (b) HWSL. (c) WS-SOD. (d) PSL. (e) Grad-CAM. (f) DeepSolar. (g) PS-CNNLC. (h) SP-RAN.

better designed network architecture, in which the RABs are integrated with FPN to handle the large intraclass variations in RSIs. Then, rather than using the simple WCE loss, a new loss function, i.e., the CA loss, is proposed, with confidence weighting introduced aiming to reduce the negative impacts of bad-quality PLs, and the gated CRF loss is also introduced to suppress the background. Finally, PS-CNNLC imposes three empirical constraints to help the label correction process, which inevitably introduces new falsely labeled pixels and mixes irrelevant surroundings with the desired objects. The proposed SP-LC strategy mitigates this problem via dynamically selecting reliable results to update the PLs in the correction process.

Both visual and quantitative comparisons are presented as follows.

1) Visual Comparison: The mapping results by different methods are shown in Figs. 5 and 6. Results are marked with true positives (yellow), true negatives (black), false positives (green), and false negatives (red) to highlight the difference between the results.

As presented in the figures, we can see that WS-SOD can cover the complete content of the objects but exhibits deficient background suppression ability, as a large part of the rooftops is mistaken as the foreground. HWSL provides a relatively accurate estimation of solar panels' layout; however, it introduces many false-positive regions (green pixels) and false-negative ones (red pixels). Grad-CAM only predicts parts of the objects and offers unsatisfactory results when there are multiple occurrences of objects in a single image. With a much shallow network architecture, PSL is prone to miss some targets, especially when it comes to small objects located in the complex background. DeepSolar has much fewer false negatives at the cost of increasing false positives. PS-CNNLC also leveraged unlabeled data and label updating, and compared with the methods aforementioned, improvements can be observed in the boundary maintenance. By contrast, despite some scattered background noises, the proposed SP-RAN demonstrates a superior ability to locate solar panels accurately and discover the complete object areas with sharp boundaries. Especially, when multiple instances appear in the image, the SP-RAN substantially outperforms other methods in producing precise mapping results.

2) Quantitative Comparison: Five evaluation metrics are used to quantitatively assess the performance of the SP-RAN, including overall accuracy, precision, recall, F_1 score, and Intersection over Union (IoU). Precision–recall curves (PRCs) are also employed to conduct a further assessment. For PRCs, varying thresholds are first utilized to transform the mapping results into binary maps, which are subsequently compared with the manually annotated ground truth to generate precision and recall.

The numerical results are summarized in Table II, with the best results reported in bold (larger values indicate better quality). PRCs for different methods are presented in Fig. 7.

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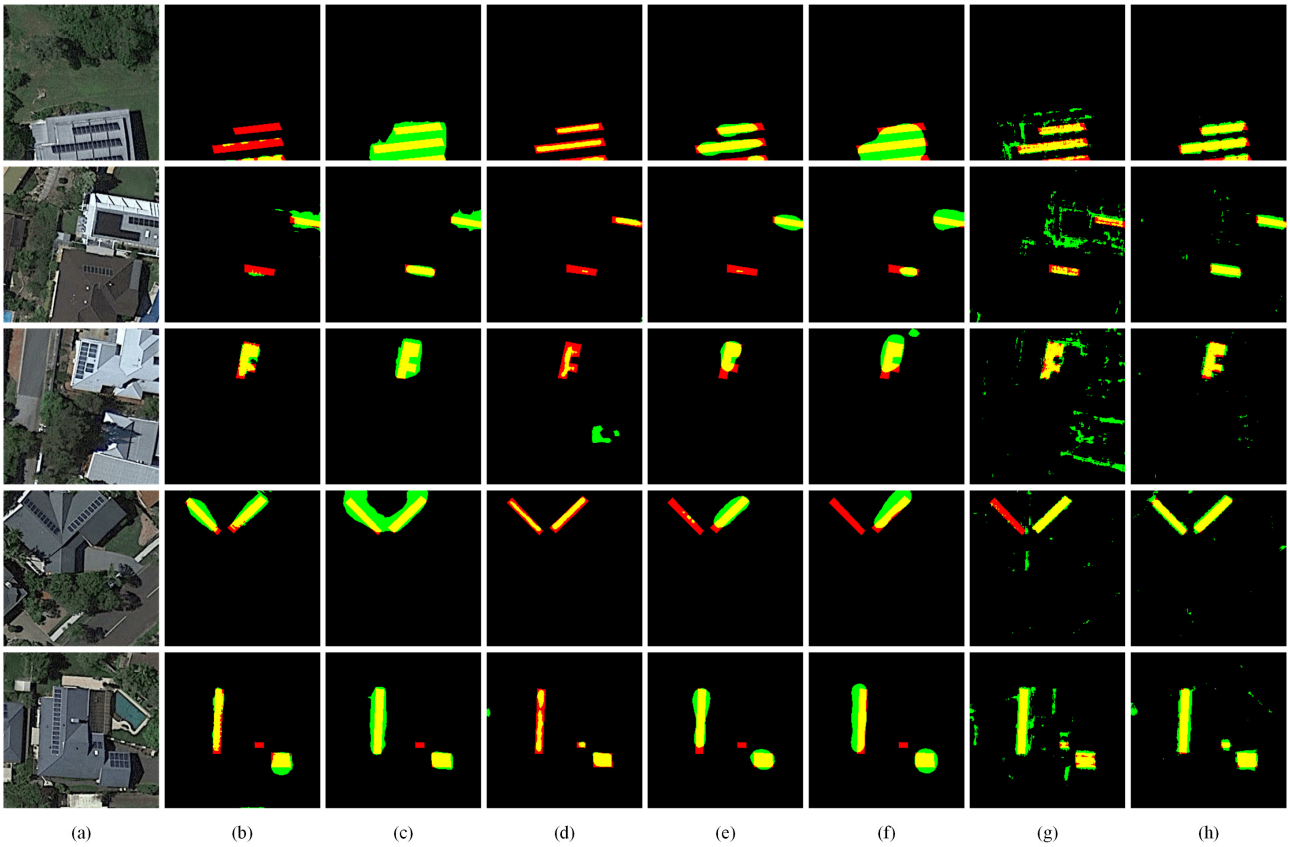


Fig. 6. Comparisons of mapping results on the GMS-BRIS dataset, marked with true positives (yellow), true negatives (black), false positives (green), and false negatives (red). (a) Original images. (b) HWSL. (c) WS-SOD. (d) PSL. (e) Grad-CAM. (f) DeepSolar. (g) PS-CNNLC. (h) SP-RAN.

TABLE II
QUANTITATIVE EVALUATION RESULTS ON THE GSM-ACT AND GSM-BRIS DATASETS

Data set	Metric	HWSL	WS-SOD	PSL	Grad-CAM	DeepSolar	PS-CNNLC	SP-RAN
GSM-ACT	*Accuracy*	97.90%	98.40%	97.96%	98.98%	97.26%	98.59%	**99.07%**
	Precision	48.29%	56.98%	49.15%	**75.55%**	40.62 %	60.70%	73.13%
	Recall	69.65%	82.60%	49.55%	72.34%	79.38%	83.93%	**84.02%**
	F_1	57.04%	67.43%	49.34%	73.91%	53.74%	70.45%	**78.20%**
	IoU	39.90%	50.87%	32.75%	58.62%	36.75%	54.38%	**64.21%**
GSM-BRIS	*Acurracy*	98.13%	97.82%	98.61%	**98.66%**	97.33%	97.68%	98.61%
	Precision	59.29%	53.49%	**87.76%**	74.45%	47.57%	52.82%	66.59%
	Recall	68.26%	**91.51%**	50.96%	69.85%	74.41%	61.05%	88.14%
	F_1	63.46%	67.52%	64.48%	72.08%	58.04%	56.63%	**75.86%**
	IoU	46.48%	50.96%	47.58%	56.35%	40.88%	39.50%	**61.11%**

From the results, we can observe that our method achieves the best overall performance with 78.20% $F1$ score and 64.21% IoU score on the GSM-ACT dataset and 75.86% $F1$ score and 61.11% IoU score on the GSM-BRIS dataset. WS-SOD exhibits high recall and relatively low precision, especially on the GSM-BRIS dataset, which is consistent with the observation in the visual comparison part. Grad-CAM has a much better precision but fails to locate multiple objects, thus exhibiting unsatisfactory performance in the recall. DeepSolar and PS-CNNLC give priority to discovering complete objects and reducing false alarms, respectively. Compared with the state-of-the-art methods aforementioned, SP-RAN achieves at least 3% improvement in F_1 score on both the GSM-ACT and GSM-BRIS datasets, which are considerable gains for the challenging weakly supervised object segmentation task.

D. Ablation Study

We also carried out experiments to investigate how the Grad-CAM at different convolutional layers, different classification networks, CA loss, unlabeled data, SP-LC strategy, and

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TABLE III
ABLATION STUDY ON THE GSM-ACT AND GSM-BRIS DATASETS

Data set	GSM-ACT					GSM-BRIS				
Metric	*Accuracy*	*Precision*	*Recall*	F_1	*IoU*	*Accuracy*	*Precision*	*Recall*	F_1	*IoU*
RAN (*baseline*)	98.35%	55.54%	**87.97%**	68.09%	51.62%	96.92%	44.37%	**95.69%**	60.63%	43.50%
w/o Confidence weighting	98.87%	68.15%	82.11%	74.48%	59.34%	98.52%	**66.93%**	79.45%	72.65%	57.05%
w/o Gated CRF regularizer	98.98%	69.97%	85.67%	77.02%	62.64%	98.48%	63.94%	89.05%	74.43%	59.28%
w/o CA loss	98.97%	**73.21%**	76.43%	74.79%	59.73%	98.39%	61.93%	90.96%	73.69%	58.34%
w/o Unlabeled data	98.98%	70.94%	83.48%	76.70%	62.21%	98.48%	63.74%	89.29%	74.39%	59.22%
w/o SP-LC	98.93%	69.00%	85.09%	76.20%	61.56%	98.34%	60.99%	91.35%	73.14%	57.66%
Proposed SP-RAN	**99.07%**	73.13%	84.02%	**78.20%**	**64.21%**	**98.61%**	66.59%	88.14%	**75.86%**	**61.11%**

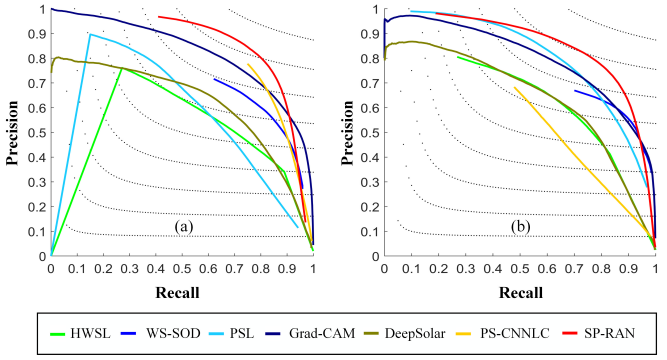


Fig. 7. PRCs of the proposed SP-RAN method and other approaches on (a) GSM-ACT and (b) GSM-BRIS datasets.

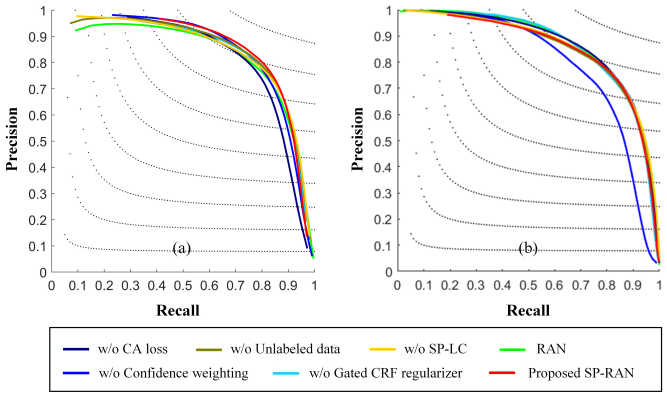


Fig. 8. PRCs of ablation study on (a) GSM-ACT and (b) GSM-BRIS datasets.

residual aggregation architecture benefit the proposed method. The results of the ablation study are reported in Tables III–V and Fig. 8. The RAN trained with the training set and WCE loss is the baseline of our work.

1) Influence of the Grad-CAM at Different Convolutional Layers: Fig. 9 shows how localizations change qualitatively as we performed Grad-CAM with respect to different feature maps in the classification network. We can see that the best localization is usually obtained with the deeper convolutional layer in the network, and Grad-CAM performs worse at shallower layers. It is noteworthy that, with the shallow layer conv3_3, Grad-CAM sometimes fails to localize the targets,

TABLE IV
QUANTITATIVE EVALUATION RESULTS WITH GRAD-CAM AT DIFFERENT LAYERS ON THE GSM-ACT AND GSM-BRIS DATASETS

Data set	Metric	w/ conv3_3	w/ conv4_3	w/ conv5_3
GSM-ACT	*Accuracy*	98.55%	**99.07%**	95.34%
	Precision	65.05%	**73.13%**	28.18%
	Recall	60.28%	84.02%	**85.42%**
	F_1	62.57%	**78.20%**	42.38%
	IoU	45.53%	**64.21%**	26.89%
GSM-BRIS	*Acurracy*	98.36%	**98.61%**	97.94%
	Precision	63.61%	**66.59%**	55.04%
	Recall	79.07%	88.14%	**93.27%**
	F_1	70.51%	**75.86%**	69.22%
	IoU	54.45%	**61.11%**	52.93%

with the activation maps corrupted by a large amount of noise; with the deeper layer conv5_3, a large proportion of rooftops are taken as the desired objects by mistake. To reveal the difference quantitatively, we generated activation maps with conv3_3, conv4_3, and conv5_3 and used the PLs to train the proposed SP-RAN. Quantitative results are shown in Table IV. We can see that on both datasets, SP-RAN with conv4_3 provides superior performance. The employment of conv5_3 helps improve the recall but leads to much worse precision.

2) Adopting Different Backbones of the Image-Level Classification Network: As Grad-CAM utilized an image-level classification network to produce activation maps as initial PLs, it is meaningful to understand the impacts of different backbones of the classification network. Hence, we replaced VGG16 with ResNet18 [60] to show the impacts. For ResNet18, we used the activation maps with Grad-CAM at the last convolutional layer to train the proposed SP-RAN; for VGG16, conv4_3 was employed in Grad-CAM and SP-RAN. Table V compares the mapping results of Grad-CAM and SP-RAN with ResNet18 and VGG16 quantitatively. The training configuration of ResNet18 is the same as that of VGG16 in our work. With different classification networks, Grad-CAM provides varying-quality activation maps: activation maps produced by Grad-CAM with VGG16 have fewer false negatives, while those produced with ResNet18 exhibit higher precision

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TABLE V
QUANTITATIVE PERFORMANCE COMPARISON WITH DIFFERENT BACKBONES OF THE CLASSIFICATION NETWORK

Data Set	Metric	Grad-CAM		SP-RAN	
		ResNet18	VGG16	ResNet18	VGG16
GMS-ACT	*Accuracy*	98.94%	98.98%	98.81%	**99.07%**
	Precision	**88.31%**	75.55%	64.55%	73.13%
	Recall	54.50%	72.34%	**89.80%**	84.02%
	F_1	67.40%	73.91%	75.18%	**78.20%**
	IoU	50.83%	58.62%	60.23%	**64.21%**
GMS-BRIS	*Acurracy*	98.54%	**98.66%**	98.19%	98.61%
	Precision	**74.91%**	74.45%	58.48%	66.59%
	Recall	61.78%	69.85%	**92.38%**	88.14%
	F_1	67.71%	72.08%	71.62%	**75.86%**
	IoU	51.19 %	56.35%	55.79%	**66.11%**

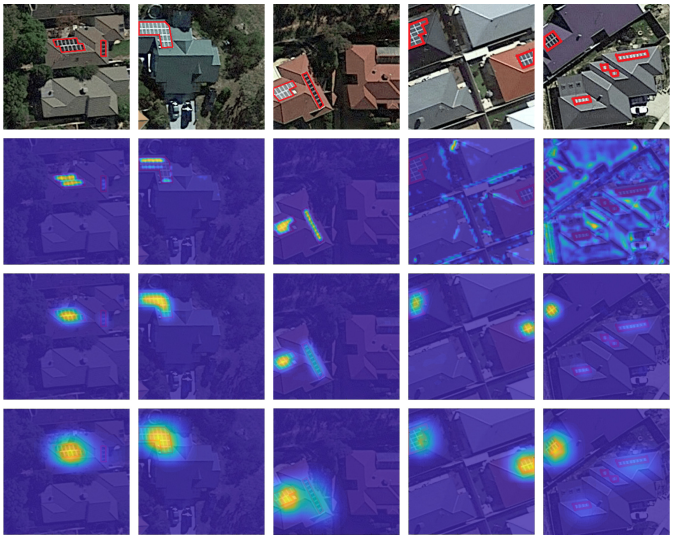


Fig. 9. Grad-CAM at different convolutional layers in the classification network. First row: original images with solar panels remarked by red lines. Second row: activation maps with conv3_3. Third row: activation maps with conv4_3. Fourth row: activation maps with conv5_3.

rates. In general, Grad-CAM with VGG16 is capable of generating initial mapping results with better overall performance. Although different backbones are employed in Grad-CAM, the proposed SP-RAN can consistently and significantly improve the F_1 score and the IoU score. Compared with ResNet18, SP-RAN with VGG16 performs better.

3) Impact of the Confidence-Aware Loss: To reveal the contribution of the CA loss, we implemented three ablation experiments to show the improvement brought by the confidence weighting and the gated CRF regularizer term. The "w/o CA loss" denotes the SP-RAN trained with only WCE loss; "w/o confidence weighting" denotes the SP-RAN trained with the WCE loss and gated CRF regularizer term [as shown in (21)]; and "w/o Gated CRF regularizer" denotes the SP-RAN trained with confidence-weighted WCE loss [as shown in (22)]

$$L = \sum_m L_{\text{wce}}^{(m)} + \lambda L_{\text{crf}}^{(m)} \tag{21}$$

$$L = \sum_m f(\mathbf{I};\Theta)_{c=1}^{(m)} \cdot L_{\text{wce}}^{(m)}. \tag{22}$$

As shown in Table III, the confidence weighting effectively increases the recall (82.11% versus 84.02% on the GSM-ACT and 79.45% versus 88.14% on the GSM-BRIS), while the Gated CRF regularizer significantly helps the model reduce false alarms and achieve higher precision (69.97% versus 73.13% on the GSM-ACT and 63.94% versus 66.59% on the GSM-BRIS). The CA loss combines the benefits of the two strategies and helps boost the overall performance, with the F_1 score increased by 3.41% and 2.17%, respectively.

4) Learning With Unlabeled Data: We utilized the unlabeled data as auxiliary information to improve the generalization ability of the model. To verify the importance of the unlabeled data, we trained RAN with the CA loss and only the training set shown in Table I, which is denoted as "w/o unlabeled data." Compared with the performance of "w/o unlabeled data", the "proposed SP-RAN" outperforms "w/o unlabeled data" with considerable increases in accuracy, F_1 score, and IoU score. On the GSM-ACT dataset, the employment of unlabeled data obtains 1.50% and 2.0% gains on the F_1 score and the IoU score; the figures are 1.47% and 1.89% for the GSM-BRIS dataset.

5) Training With the SP-LC Strategy: To reveal the effectiveness of the proposed SP-LC strategy, we further carried out an ablation experiment without the SP-LC strategy, which is shown as "w/o SP-LC" in Table III. It is noticeable that the application of the SP-LC strategy improves the precision remarkably (4.13% and 5.6% on GSM-ACT and GSM-BRIS datasets, respectively) at the cost of a slight decline in recall (1.07% and 3.21% on GSM-ACT and GSM-BRIS datasets, respectively). In general, SP-LC further boosts the performance by 2% and 2.72% F_1 scores on GSM-ACT and GSM-BRIS datasets, respectively.

6) Contribution of RAN: Since RAN is composed of two main parts, stacked RABs as the encoder path and FPN as the backbone, we added some comparative experiments to understand how much each of them contributes to the performance improvements. To reveal the contribution of RABs in the encoder path, we replaced the proposed target mapping network with the plain FPN architecture. The performance is denoted as FPN. For the FPN part, we introduced some FPN variants, including PANet [61], BiFPN [62], and NAS-FPN [63], to replace the FPN in the proposed RAN. For a fair comparison, we retained the encoder path consisting of four consecutive RABs to generate the pyramidal features. The CA loss and the SP-LC strategy are also applied in the training phase. In BiFPN and NAS-FPN, we only used one repeated block. As the extra bottom-up pathway in PANet, BiFPN, and NAS-FPN cannot produce full-resolution results for dense prediction tasks, such as object segmentation, we added a simple design to merge the information from all levels of the produced feature pyramid into a single output as in [64]. In the following context, we use RAB-PANet, RAB-BiFPN, and RAB-NAS-FPN to denote the network architectures aforementioned. Compared with the FPN architecture, PANet, BiFPN, and NAS-FPN add an extra bottom-up pathway to further enhance the multiscale feature fusion by exploring cross-scale

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TABLE VI
QUANTITATIVE PERFORMANCE COMPARISON WITH DIFFERENT BACKBONES ON GSM-ACT AND GSM-BRIS DATASETS

Method	GSM-ACT					GSM-BRIS				
	Accuracy	*Precision*	*Recall*	F_1	*IoU*	*Accuracy*	*Precision*	*Recall*	F_1	*IoU*
FPN	98.21%	53.51%	80.44%	64.27%	47.35%	97.87%	54.05%	93.85%	68.60%	52.20%
RAB-PANet	97.81%	47.42%	83.99%	60.62%	43.49%	97.53%	50.09%	**93.93%**	65.34%	48.52%
RAB-BiFPN	98.06%	51.03%	82.29%	62.99%	45.98%	97.00%	44.96%	93.07%	60.63%	43.51%
RAB-NAS-FPN	97.46%	42.73%	79.01%	55.47%	38.38%	97.15%	46.26%	91.65%	61.49%	44.39%
RAN	**99.07%**	**73.13%**	**84.02%**	**78.20%**	**64.21%**	**98.61%**	**66.59%**	88.14%	**75.86%**	**61.11%**

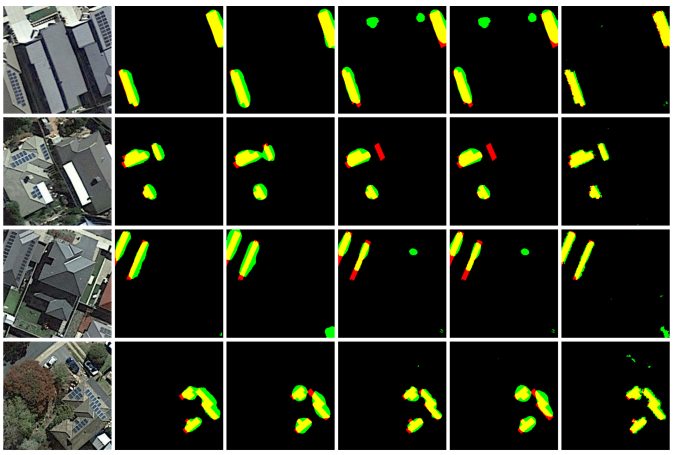


Fig. 10. Mapping results with different backbones on the GMS-ACT dataset. (a) Original images. (b) FPN. (c) RAB-PANet. (d) RAB-BiPAN. (e) RAB-NAS-FPN. (f) RAN.

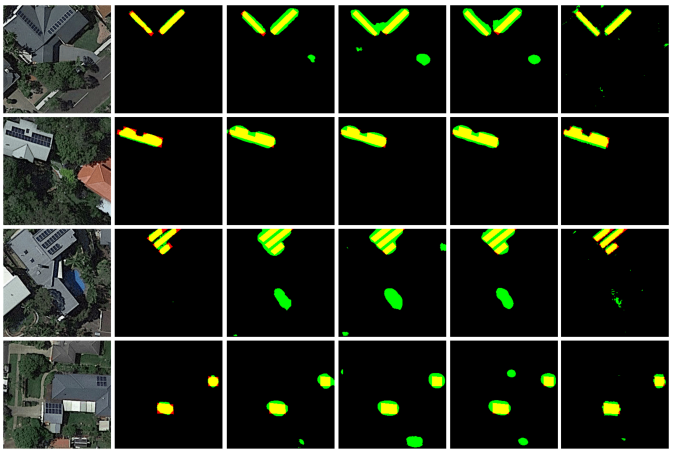


Fig. 11. Mapping results with different backbones on the GMS-BRIS dataset. (a) Original images. (b) FPN. (c) RAB-PANet. (d) RAB-BiPAN. (e) RAB-NAS-FPN. (f) RAN.

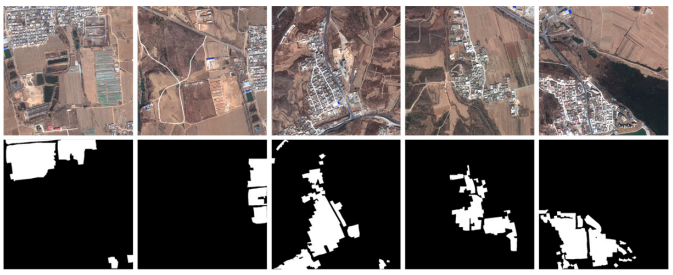


Fig. 12. Examples of images and the corresponding ground truths from the GeoEye-1 dataset. First row: RSIs. Second row: corresponding ground truths (only used for evaluation).

lateral connections. From Table VI, the comparison between RAN and FPN can substantially demonstrate the advantages of the residual aggregated architecture, i.e., RABs. As illustrated in Figs. 10 and 11, compared with RAB-PANet, RAB-BiFPN, and RAB-NAS-FPN, the proposed RAN achieves better accuracy in localizing the targets with varying appearance and has a stronger ability to cope with interference, including shadows and occlusions. RAN also provides results much closer to the actual object boundary. From Table VI, we can find that, by adopting FPN as the backbone, the proposed RAN outperforms the others by a wide margin in the overall performance.

To validate the effectiveness of RAN in other cases, we reported the performance of RAN, RAB-PANet, RAB-BiFPN, and RAB-NAS-FPN on a remote sensing dataset from the GeoEye-1 satellite. This dataset is collected for salient object detection, which takes residential areas as the target. The detailed configuration of the GeoEye-1 dataset can be found in [45]. Examples of images and the corresponding grounds truths are shown in Fig. 12. We compared the performance of RAB-PANet, RAB-BiFPN, RAB-NAS-FPN, and the proposed RAN visually and quantitatively. Corresponding results are shown in Fig. 13 and Table VII. From Fig. 13, we can observe that RAB-PANet and RAB-BiFPN show similar visual performance: they can not only discover the majority of the residential areas but also introduce a large number of false negatives and false positives. RAB-NAS-FPN suffers from a low precision rate. By contrast, despite some scattering background interference, the proposed RAN provides mapping results with less falsely predicted pixels and a more accurate object boundary. The quantitative comparison in Table VII also indicates that the proposed RAN outperforms the other methods in precision, $F1$ score, and IoU score.

Compared with the FPN architecture, PANet, BiFPN, and NAS-FPN aim to discover more effective multiscale feature fusion by exploring cross-scale lateral connections. FPN only has one bottom-up path and one top-down path, with skip connections fusing the low- and high-level feature maps. PANet, BiFPN, and NAS-FPN add an extra bottom-up pathway to further enhance the feature fusion. With more model parameters and cross-scale fusion, such a design can improve

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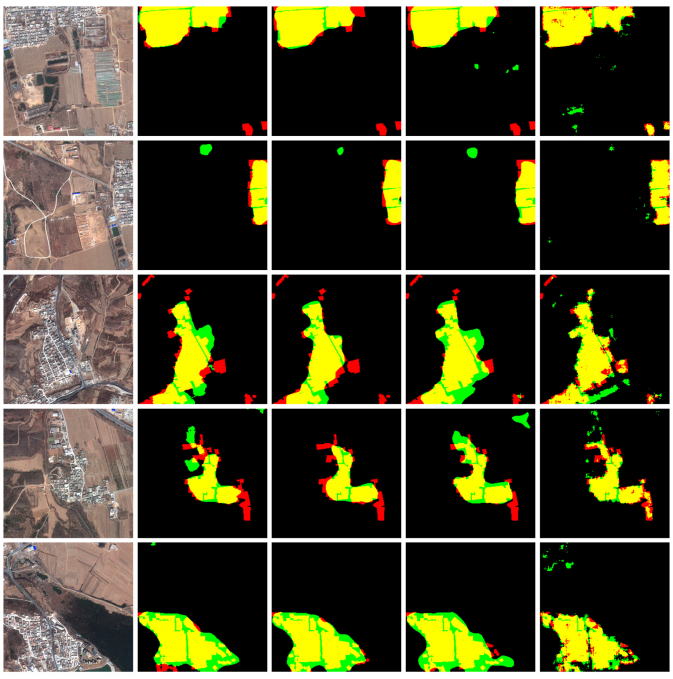


Fig. 13. Mapping results with different backbones on the GeoEye-1 dataset. (a) Original images. (b) RAB-PANet. (c) RAB-BiPAN. (d) RAB-NAS-FPN. (e) RAN.

TABLE VII
QUANTITATIVE PERFORMANCE COMPARISON WITH DIFFERENT BACKBONES ON THE GEOEYE-1 DATASET

Methods	RAB-PANet	RAB-BiFPN	RAB-NAS-FPN	RAN
Accuracy	93.71%	94.28%	92.72%	**95.02%**
Precision	74.36%	77.12%	68.69%	**80.10%**
Recall	82.88%	83.08%	**86.55%**	84.89%
F_1	78.39%	79.99%	76.59%	**82.42%**
IoU	64.46%	66.65%	62.07%	**70.10%**

the model capacity greatly. This is a large advantage when it comes to fully supervised tasks with a large number of accurate pixelwise annotations. Weakly supervised methods, however, may not benefit from such design as it can possibly lead to the overfitting problem at an early stage in the training phase, which means that it will be increasingly difficult to maintain the precise object boundary.

IV. CONCLUSION

In this article, we proposed an effective method for weakly supervised object extraction and applied it to the solar panel mapping in aerial images. With the CA loss and SP-LC strategy, the proposed method emphasizes the contribution of the high-quality PLs to the model training and minimizes the negative impacts of the bad-quality ones. The CA loss employs the classification score to adjust the contribution of training samples with quality-inconsistent PLs adaptively. Then, the SP-LC selects training samples with certain learning difficulties, which matches the current model capacity and updates their PLs instead of updating the entire set. Moreover, we proposed an RAN for target mapping, which aims at capturing complete objects with precise boundaries. Extensive experiments and the ablation study on two aerial datasets demonstrate the advantages of the proposed method in solar panel mapping. We also applied the proposed method to a remote sensing dataset from the GeoEye-1 satellite, which also indicates the effectiveness of the proposed SP-RAN.

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